

CoinGecko: Positions itself to launch corporate and institutional digital currency data services with BigQuery

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ABOUT COINGECKO



CoinGecko

About CoinGecko

With BigQuery, Co
can capture and
digital currency d
needed to meet c
and institutional
analyze historical
decision-making

CoinGecko provides a fundamental analysis of the digital currency market. In addition to tracking price, volume, and market capitalization, CoinGecko tracks community growth, open source code development, major events, and on-chain metrics.

Industries: Financial Services &
Insurance

Google Cloud results

- Supports the delivery of historical analytics services to institutional and corporate customers
- Enables business to store data at a cost 40% lower than alternative services
- Allows operation of scalable blockchain validator nodes 30% cheaper than other services
- Allows future expansion and scalability

Location: Malaysia

Supports more than 100 primary use cases about moving digital currencies

Google Cloud (<https://cloud.google.com/>)

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BigQuery (<https://cloud.google.com/bigquery/>)

Compute Engine

(<https://cloud.google.com/compute/>)

**through
Cloud
Bigtable
and Google
Kubernetes
Engine**

As digital currencies proliferate and mature, the range and quality of services that support users is increasing. Founded in 2014 and headquartered in Kuala Lumpur, Malaysia, CoinGecko

(<https://www.coingecko.com/en>) provides a digital currency market tracking website and app. The website started by ranking digital currencies by fundamental metrics such as developer activity, community size, and liquidity. “We benchmark digital currencies such as Bitcoin, Ethereum, and 6,000+ other cryptocurrencies,” says T.M. Lee, co-founder of CoinGecko. “We do this by collecting data to quantitatively and qualitatively rank the potential of each coin.

“Our aim is to become the most comprehensive data provider for the open financial market—which we believe will be facilitated by blockchain.”

As well as tracking about 6,120 coins and 399 exchanges, CoinGecko aggregates news content from a range of sources to provide a holistic service to users. “The data we provide includes price, volume, and market capitalization, while we also aim to aggregate data such as the number of developers who are working on a cryptocurrency project and

how involved and active the community of each project is on social media,” says Lee.



50 million page views per month

The CoinGecko website attracts around 50 million page views per month, while users have collectively installed the app—available since the end of 2018—about 100,000 times.

The CoinGecko website and app primarily aim to provide content for consumers. However, the business’s product is attracting increasing interest from institutions and large corporates. “We started with and we’re good at providing consumer-grade services,” says Lee. “However, we are increasingly receiving requests from corporates and institutions for solutions that meet their needs.”

These in-demand solutions included access to historical data to analyze trends and movements and make decisions about digital currency activities.

For example, funds could use the data to perform strategy backtesting—using historical data to reconstruct trades that would have occurred had certain strategies been applied—or calculating portfolio returns. Developers can leverage the data to develop trading bots and stock screener, general stock chart, and stock pricing apps. “Our 11-person business does not have the capacity to deliver those solutions yet—but we plan to do so in the near future,” says Lee.

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—T.M. Lee, co-founder,
CoinGecko

Capturing large volumes of data

However, delivering those solutions involved capturing, storing, and archiving large volumes of data, including price “ticks”—minimum upward or downward movement—of all the digital coins being traded. “We could not keep the raw data corporates and institutions needed,” says Lee. “The cost of a ‘hot’ relational database is too high to be storing large volumes of rarely accessed historical data.”

BigQuery provides the answer

CoinGecko reviewed its options for data management and archiving and selected the [BigQuery](#) (/bigquery) analytics data warehouse. “We primarily chose BigQuery because it gave us the ability to store data cost-effectively in an accessible location,” says Lee. “In addition, the fact BigQuery provided access to a dataset for Ethereum—a prominent digital currency that has captured the

attention of technologists, financiers and economists—also helped our decision-making.”

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—T.M. Lee, co-founder,
CoinGecko

CoinGecko is now using BigQuery to warehouse more than four TB of primary data, while replication grows the total volume of data stored to more than 12 TB. “Moving to BigQuery has enabled us to keep our storage costs 40% lower than on other services while retaining data in an easily accessible location,” says Lee.

The fact that BigQuery provides access to datasets for Ethereum and Bitcoin also enables CoinGecko to avoid the cost of running a node and associated compute and storage to capture transaction data and perform queries. “The costs we do not incur also include training engineers to learn, set up, and maintain nodes, and indexing blockchain datasets for analysis,” says Lee. “With BigQuery, we can simply jump right into consuming blockchain datasets for research.”

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Using Compute Engine to run validation services for blockchain networks

The business is also using virtual machine instances to operate validation services for some blockchain and digital currency networks. “We are running validator nodes through Compute Engine (/compute), mainly for Delegated Proof of Stake consensus algorithm blockchain networks such as Steem, IoTeX, and Tomochain,” says Lee. “The role of a validator is to secure a blockchain network by verifying and validating transactions and participating in governance activities when necessary.”

“The per-instance cost through Compute Engine is 30% cheaper than the alternatives, while Google Cloud provides proactive advice when we over-provision an instance, allowing us to dial back to a more appropriate, cost-effective level.”

With BigQuery and Compute Engine running smoothly and meeting the business’s requirements, CoinGecko is looking at other Google Cloud services

to help future-proof its infrastructure and further control its costs. “We imagine a world where more and more assets will be tokenized on the blockchain,” says Lee. “To make sense of and index that data, we need to be prepared with services such as [Cloud Bigtable](#) (/bigtable) to provide a petabyte scale, fully managed NoSQL database service and [Google Kubernetes Engine](#) (/kubernetes-engine) to provide a scalable, managed service to deploy containerized applications.”

“As we grow our use of Google Cloud over time, we expect to see a lot more nice features, and we believe there is a lot in this family of services that can add value to our business,” concludes Lee.

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